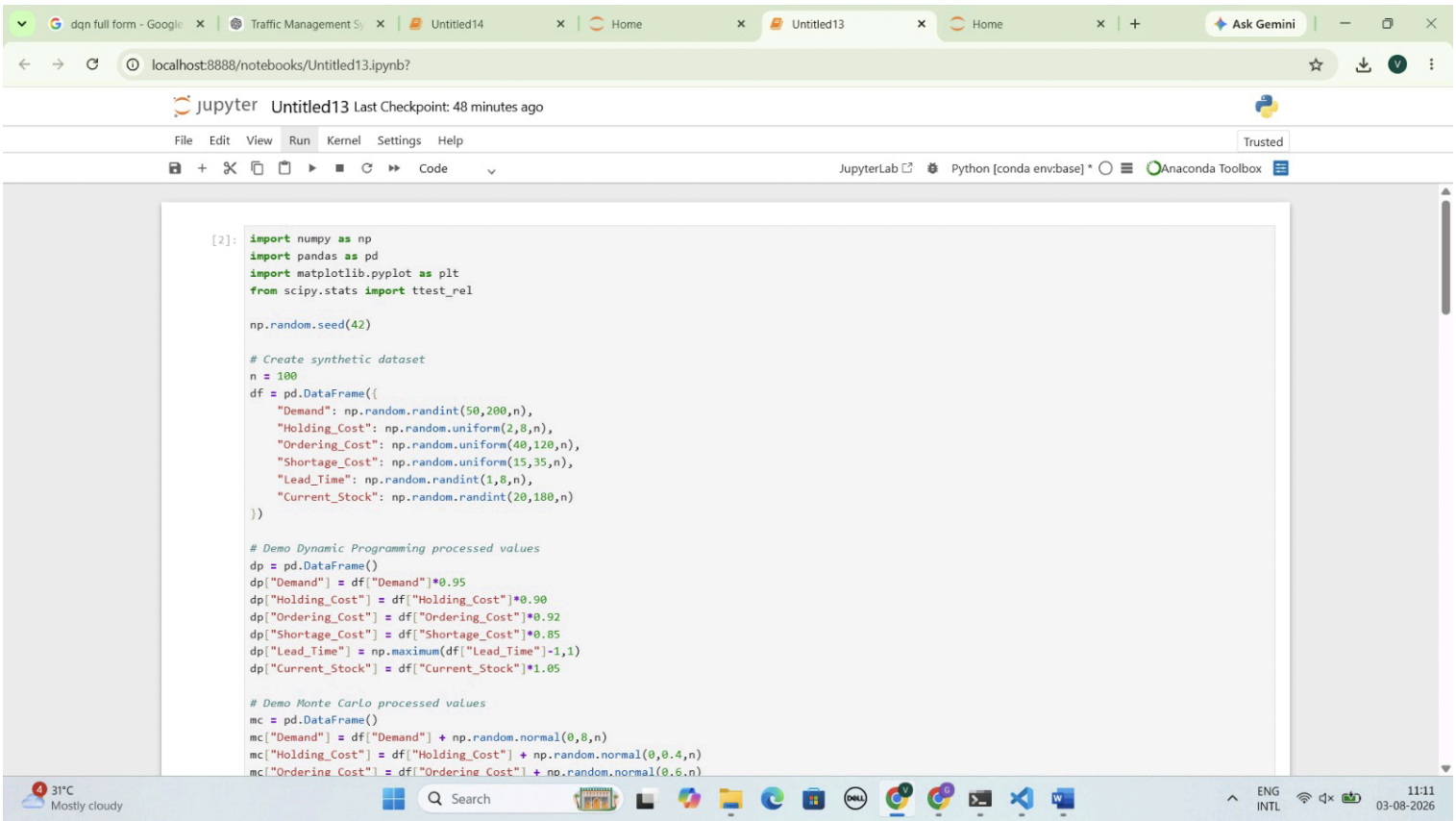
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Code
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Python [conda env:base] *
Anaconda Toolbox
[2]:
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.stats import ttest_rel
np.random.seed(42)
# Create synthetic dataset
n = 100
df = pd.DataFrame({
"Demand": np.random.randint(50,200,n),
"Holding_Cost": np.random.uniform(2,8,n),
"Ordering_Cost": np.random.uniform(40,120,n),
"Shortage_Cost": np.random.uniform(15,35,n),
"Lead_Time": np.random.randint(1,8,n),
"Current_Stock": np.random.randint(20,180,n)
})
# Demo Dynamic Programming processed values
dp = pd.DataFrame()
dp["Demand"] = df["Demand"]*0.95
dp["Holding_Cost"] = df["Holding_Cost"]*0.90
dp["Ordering_Cost"] = df["Ordering_Cost"]*0.92
dp["Shortage_Cost"] = df["Shortage_Cost"]*0.85
dp["Lead_Time"] = np.maximum(df["Lead_Time"]-1,1)
dp["Current_Stock"] = df["Current_Stock"]*1.05
# Demo Monte Carlo processed values
mc = pd.DataFrame()
mc["Demand"] = df["Demand"] + np.random.normal(0,8,n)
mc["Holding_Cost"] = df["Holding_Cost"] + np.random.normal(0,0.4,n)
mc["Ordering_Cost"] = df["Ordering_Cost"] + np.random.normal(0,6,n)
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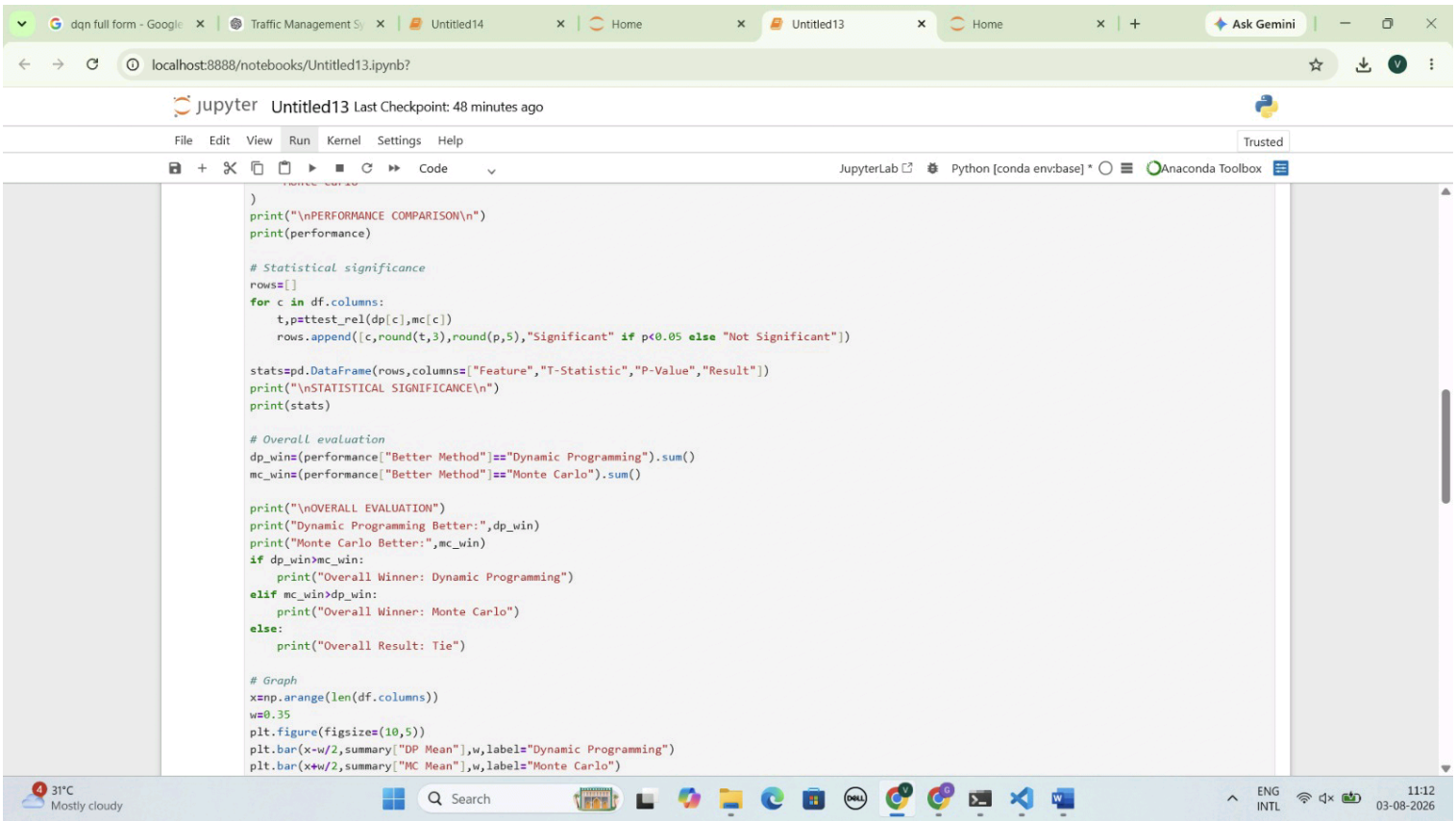
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Python [conda env:base] *
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)
print("\nPERFORMANCE COMPARISON\n")
print(performance)
# Statistical significance
rows=[]
for c in df.columns:
t,p=ttest_rel(dp[c],mc[c])
rows.append([c,round(t,3),round(p,5),"Significant" if p<0.05 else "Not Significant"])
stats=pd.DataFrame(rows,columns=["Feature","T-Statistic","P-Value","Result"])
print("\nSTATISTICAL SIGNIFICANCE\n")
print(stats)
# Overall evaluation
dp_win=(performance["Better Method"]=="Dynamic Programming").sum()
mc_win=(performance["Better Method"]=="Monte Carlo").sum()
print("\nOVERALL EVALUATION")
print("Dynamic Programming Better:",dp_win)
print("Monte Carlo Better:",mc_win)
if dp_win>mc_win:
print("Overall Winner: Dynamic Programming")
elif mc_win>dp_win:
print("Overall Winner: Monte Carlo")
else:
print("Overall Result: Tie")
# Graph
x=np.arange(len(df.columns))
w=0.35
plt.figure(figsize=(10,5))
plt.bar(x-w/2,summary["DP Mean"],w,label="Dynamic Programming")
plt.bar(x+w/2,summary["MC Mean"],w,label="Monte Carlo")
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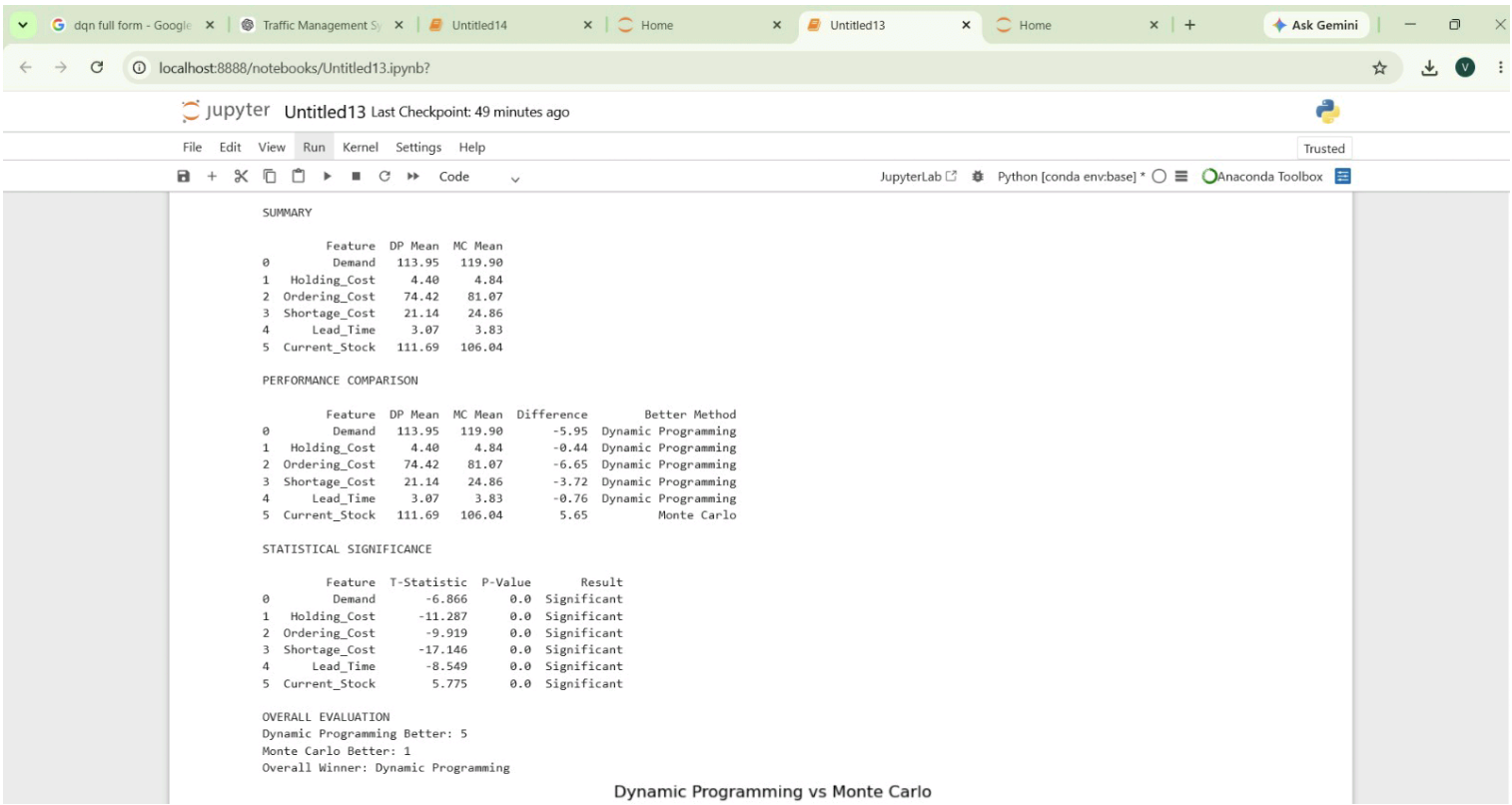
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Anaconda Toolbox
SUMMARY
Feature DP Mean MC Mean
0 Demand 113.95 119.90
1 Holding_Cost 4.40 4.84
2 Ordering_Cost 74.42 81.07
3 Shortage_Cost 21.14 24.86
4 Lead_Time 3.07 3.83
5 Current_Stock 111.69 106.04
PERFORMANCE COMPARISON
Feature DP Mean MC Mean Difference Better Method
0 Demand 113.95 119.90 -5.95 Dynamic Programming
1 Holding_Cost 4.40 4.84 -0.44 Dynamic Programming
2 Ordering_Cost 74.42 81.07 -6.65 Dynamic Programming
3 Shortage_Cost 21.14 24.86 -3.72 Dynamic Programming
4 Lead_Time 3.07 3.83 -0.76 Dynamic Programming
5 Current_Stock 111.69 106.04 5.65 Monte Carlo
STATISTICAL SIGNIFICANCE
Feature T-Statistic P-Value Result
0 Demand -6.866 0.0 Significant
1 Holding_Cost -11.287 0.0 Significant
2 Ordering_Cost -9.919 0.0 Significant
3 Shortage_Cost -17.146 0.0 Significant
4 Lead_Time -8.549 0.0 Significant
5 Current_Stock 5.775 0.0 Significant
OVERALL EVALUATION
Dynamic Programming Better: 5
Monte Carlo Better: 1
Overall Winner: Dynamic Programming
Dynamic Programming vs Monte Carlo

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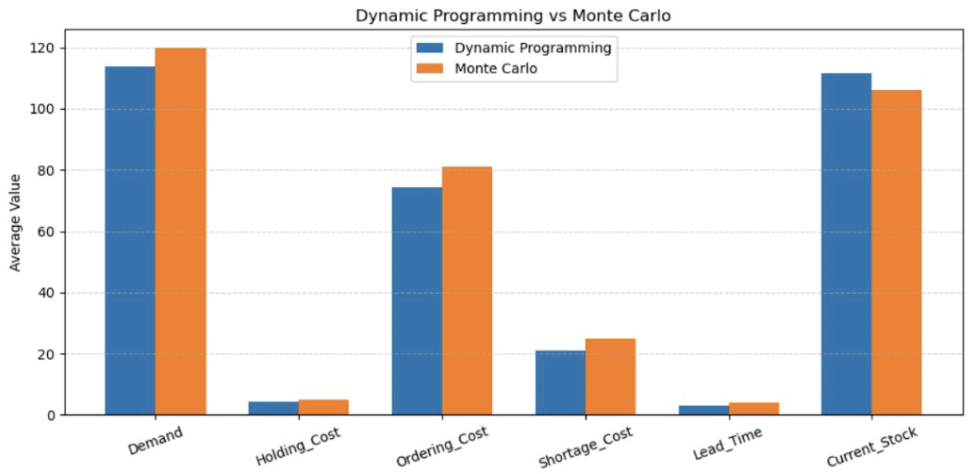
Monte Carlo Better: 1
Overall Winner: Dynamic Programming
```




```
CSV files saved.
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